

RESEARCH EXPERIENCE

My research journey began with Domain Adaptive Object Detection, which combines two core concepts: **perception and domain transfer**. Interestingly, this was not only the starting point of my work, but the underlying philosophy has profoundly shaped all of my subsequent research. I firmly believe that breakthroughs in any field rely on two pillars: **the depth of perception within that field and the effective transfer of insights from other fields**. This philosophy has led my research to span a remarkably broad range of topics, covering four main areas:

- AI for Human Life 2025 - Present
 - Video Diffusion Model; 3D Spatial Reasoning
- AI for Autonomous Mobility 2020 - Present
 - 3D Occupancy Prediction; Open-vocabulary/Cross-domain/Open-set Object Detection
- AI for Scientific Innovation 2024 - 2025
 - Nanophotonics; Meta Optics; Graph-based Learning
- AI for Transforming Medicine 2020 - 2024
 - MRI Brain Analysis; Medical Report Generation; 4D Surgical Simulation

PERSONAL HIGHLIGHT

- Academic & Industrial Insight
 - Bridging top-tier research and entrepreneurship, while founding an HK startup that secured seed-round funding.
- Large-Scale GPU Experimentation
 - Extensive hands-on experience using large-scale GPU clusters (10,000+ GH200).
- Leadership & Collaboration
 - Led Swiss national AI initiative with 15 top Swiss PIs (e.g., Marc Pollefeys @ ETH); NeurIPS workshop organizer.
- High-Quality First-Author Output
 - 10+ first-author papers at top venues; A single-project codebase attracted 700+ GitHub stars within two months.

RESEARCH HIGHLIGHT

- Selected as Oral/Spotlight/Highlight Presentation (1st Author)
 - AAAI 2022; CVPR 2022; ICCV 2023; ECCV 2024; ICCV 2025; NeurIPS 2025
- Selected as Best Paper Finalist (1st Author)
 - CVPR 2022 (33/8161)
- Full Score Submission (1st Author)
 - ICCV 2025, Final Score: 6, 6, 6
- Selected as Cover Paper (2nd Author)
 - ACS Photonics, 2024

WORK EXPERIENCE

- École Polytechnique Fédérale de Lausanne (EPFL) Feb. 2025 - Present
 - Research Scientist, Robotics and Computer Vision
 - Supervisor: Prof. Alexandre Alahi
- The Chinese University of Hong Kong (CUHK) Mar. 2024 - Feb. 2025
 - Postdoc, Electronic Engineering
 - Supervisor: Prof. Yixuan Yuan

EDUCATION

- City University of Hong Kong (CityU) Sep. 2020 - Feb. 2024
 - Ph.D., Electrical Engineering
 - 3.5-Year; Early Graduation
 - Supervisor: Prof. Yixuan Yuan
- Tianjin University (TJU) Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA: 3.83/4.00 (Top 2%)

1. **[Video Generation]** Stable Video Infinity: Infinite-Length Video Generation with Error Recycling
Wuyang Li, Wentao Pan, Po-Chien Luan, Yang Gao, Alexandre Alahi
 ArXiv Preprint, 2025
2. **[3D Perception]** VoxDet: Rethinking 3D Semantic Occupancy Prediction as Dense Object Detection
Wuyang Li, Zhu Yu, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2025, **Spotlight (3%)**
3. **[AI for Science]** MetaScope: Optics-Driven Neural Network for Ultra-Micro Metalens Endoscopy
Wuyang Li*, Wentao Pan*, Xiaoyuan Liu*, Zhendong Luo, Chenxin Li, Hengyu Liu, Din Ping Tsai, Mu Ku Chen, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2025, **Full scores in the final rating (6, 6, 6)**
4. **[Backbone]** U-kan makes strong backbone for medical image segmentation and generation
 Chenxin Li*, Xinyu Liu*, **Wuyang Li***, Cheng Wang*, Hengyu Liu, Yifan Liu, Zhen Chen, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence, (AAAI), 2025, **AAAI 2025 Top-1 Influential Paper**
5. **[Diffusion]** CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
 European Conference on Computer Vision (ECCV), 2024, **Oral (1.8%)**
6. **[Video Generation]** Boosting Medical Image Analysis via Motion-Informed Generative Videos
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
7. **[Open-World Learning]** Novel Scenes & Classes: Towards Adaptive Open-set Object Detection
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2023, **Oral (1.8%)**
8. **[Open-World Learning]** Adjustment and Alignment for Unbiased Open Set Domain Adaptation
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
9. **[Domain Shift]** SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023
10. **[Domain Shift]** SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, **Best Paper Finalist (0.4%)**
11. **[Domain Shift]** SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence (AAAI), 2022, **Oral (4.5%)**
12. **[Brain Analysis]** Synthesizing Realistic fMRI: A Physiological Dynamics-Driven Hierarchical Diffusion Model for Efficient fMRI Acquisition
 Yufan Hu, Yu Jiang, **Wuyang Li**, Yixuan Yuan
 International Conference on Learning Representations (ICLR), 2025
13. **[Brain Analysis]** InstantSplamp: Fast and Generalizable Stenography Framework for Generative Gaussian Splatting
 Chenxin Li*, Hengyu Liu*, Zhiwen Fan, **Wuyang Li**, Yifan Liu, Panwang Pan, Yixuan Yuan
 International Conference on Learning Representations (ICLR), 2025
14. **[Backbone]** Universal Domain Adaptive Object Detection via Dual Probabilistic Alignment
 Yufanfan Zheng*, Jinlin Wu*, **Wuyang Li**, Zhen Chen
 The Association for the Advance of Artificial Intelligence, (AAAI), 2025
15. **[AI for Science]** Stereo Vision Meta-Lens-Assisted Driving Vision
 Xiaoyuan Liu, **Wuyang Li**, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
 ACS Photonics, 2024, **Selected as Cover Paper**
16. **[SAM]** Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM
 Chenxin Li*, Yuzhi Huang*, **Wuyang Li**, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
 Conference on Neural Information Processing Systems (NeurIPS), 2024

17. **[SAM] When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
Yifan Liu, Wuyang Li, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, **Early Accept (14.0%)**
18. **[Diffusion] Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
19. **[Brain] FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
20. **[3D] LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
21. **[Video Generation] Endora: Video Generation Models as Endoscopy Simulators**
Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
22. **[Brain] Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (TMI), 2024
23. **[Multi-Modality] MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
Qiushi Yang, Wuyang Li, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (ICCV), 2023
24. **[Noisy Labels] Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, **Oral, Early Accept**
25. **[Noisy Labels] Towards Robust Adaptive Object Detection under Noisy Annotations**
Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuann
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
26. **[Object Detection] HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (TIP), 2021

AWARDS

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| <ul style="list-style-type: none"> • Outstanding Academic Performance Award, CityU • Outstanding research achievements • Research Tuition Scholarship, CityU • Outstanding research achievements • Ranked First in Polyp Detection Challenge • 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021) • Undergraduate National Scholarship • The highest undergraduate honor for a student in China • Silver Award in Tianjin Advanced Mathematical Competition • As a selected participant representing Tianjin University | <div>2022 - 2023</div> <div>2022 - 2023</div> <div>2021</div> <div>2017 - 2018</div> <div>2017</div> |
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SERVICES

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| <ul style="list-style-type: none"> • Leader of Large Swiss AI Initiative Project [Link] • Led a large national project with 15 top Swiss PIs; Large-scale GPU clusters experience (10,000+ GH200). • NeruIPS-24 Workshop Organizer [Link] • Organized the Advancements in Multi-modal Medical Foundation Models workshop • Research Funding Proposal • Assisted my advisor in completing several research funding proposals • Startup Funding [Link] • Successfully secured seed funding from HK Tech 300 and HKSTP • Founder of ScholaGO Education Technology Company Limited, Hong Kong [Link] • Founded a company to develop a multi-modal educational product • Conference and Journal Reviewer • ICLR-25; NeurIPS-25/24 (Top Reviewer); ICML 2025; CVPR-23/24; ICCV-23/25; T-PAMI; IJCV; TIP | <div>2025</div> <div>2024</div> <div>2022 - 2024</div> <div>2023</div> <div>2024</div> |
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